

Molecular Dynamics Simulations of Heterogeneous Contaminant Films

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ABSTRACT

In this paper, we discuss a recent effort on utilizing molecular dynamics (MD) simulations to investigate the buildup and desorption of heterogeneous contaminant films on a gold substrate in a space environment. Our simulation models a Quartz-Crystal Microbalance (QCM) crystal exposed to ambient molecular nitrogen, water vapor, and several small hydrocarbons. A Berendsen thermostat in LAMMPS is used to implement a time-variant temperature ramp from the initial substrate temperature of 73 K up to temperatures of 450 K, mimicking desorption during thermal events. Our primary interest is in exploring the interactions between desorbing water molecules and hydrocarbon contaminants. Results from this study will eventually be used to derive a reduced-order model compatible with three-dimensional molecular transport codes. Our MD simulations show that the molecular layers dynamically rearrange such that species tend to desorb in order of increasing activation energy, with more weakly bound species desorbing first and more strongly bound species remaining longer. Importantly, the simulations also reveal that the same molecular species can exhibit different desorption behavior depending on the underlying surface to which they are attached, with hydrocarbons directly adsorbed on gold remaining well after similar molecules on water layers have fully desorbed. These findings underscore the importance of accounting for both molecular binding strength and source-target material pairings in spacecraft contamination modeling.

Keywords: Molecular Dynamics, spacecraft contamination

1. INTRODUCTION

Understanding the formation and evolution of spacecraft contamination films at the atomic level requires models that go beyond traditional assumptions about surface uniformity and molecular behavior. Classical adsorption models like the Langmuir adsorption model and BET theory are two of the most common in surface science but are fundamentally limited in several key respects. Both assume a uniform substrate with fixed, identical binding sites—meaning they do not account for different source–target material pairings and activation energies, among other effects. Langmuir¹ theory captures monolayer adsorption with no multilayer interactions, while BET² extends this to allow for multilayer buildup. However, neither model incorporates molecular identity, intermolecular interactions, or dynamic restructuring of heterogeneous films over time.

Likewise, the temperature-dependent sticking coefficient approach—which is a more recent approach used in the spacecraft contamination community³—typically assumes a single temperature-dependent sticking coefficient for each species based on its direct interaction with the exposed substrate. This simplification does not account for complex layering effects or re-emission from intermediate surfaces (e.g., desorption of a molecule from a water layer rather than directly from gold). As a result, such models can miss critical behaviors observed in more physically detailed simulations or experiments, especially when multi-species films evolve over time and temperature. And even more recent multi-species models^{4,5} which have gained popularity over the last 2-4 years still don’t take into account many of these source-target material pairings or dynamic restructuring effects.

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These limitations are particularly important when simulating real-world spacecraft conditions. Most spacecraft undergo regular thermal cycling as they orbit Earth (moving in and out of eclipse), or during mission-specific transitions—such as JWST warming up after sunshield deployment or Dragonfly descending through Titan’s dense atmosphere. In cryogenic environments, surface temperatures may fluctuate between 70 K and 300 K. Under these conditions, spacecraft surfaces are expected to be coated in thin films composed of water, hydrocarbons, nitrogen, and other molecular species which comprise the most common spacecraft outgassing constituents. However, how these species interact—particularly as temperature ramps up and one begins to desorb—is not well understood. For instance, water may “boil off” at relatively low temperatures, leaving behind hydrocarbon films that were previously layered on top of it (or the hydrocarbons may leave first, trap the water, or even be “carried away” early by the water). The way in which these species separate, reconfigure, or persist can have significant implications for contamination control and modeling.

In this work, we apply molecular dynamics (MD) simulations to better understand the behavior of these heterogeneous films under thermal ramping. Specifically, we examine mixtures of water, methane, nitrogen, decane, and toluene on a gold substrate—chosen since the surfaces of Quartz Crystal Microbalances (QCMs) - the surface considered in this work - are coated in gold. QCMs are highly sensitive instruments used to measure small mass changes on a quartz crystal sensor by taking the difference of the frequencies of that crystal and a reference quartz crystal inside the sensor. QCMs are particularly useful for applications requiring precise monitoring of thin films and surface interactions (hence why they’re commonly used in spacecraft contamination testing). We begin the paper by describing the numerical model. We then simulate the deposition of these molecules at a thermal velocity corresponding to 273.15 K (i.e. 0 °C) onto a substrate held at 73 K (i.e. -200.15 °C). We next consider thermal desorption of these films by applying temperature ramps from the original 73 K substrate set point to 150 K, 250 K, 350 K, and 450 K, respectively. Our results show that more weakly bound species tend to desorb first while the more strongly bound molecules with higher activation energies remain behind. Interestingly, in our simulations, we found that the same molecular type can exhibit different desorption behaviors depending on whether it was initially deposited on water or on gold, underscoring that there’s a strong influence of source-target pairings on activation energy. These effects are largely absent from classical or semi-empirical models but to our knowledge, are explored for the first time (in a spacecraft contamination context) using MD simulations. This paper aims to explore and quantify these dynamics and lay the foundation for more accurate, physics-based reduced-order contamination models.

2. NUMERICAL MODEL

In this paper, we apply molecular dynamics (MD) simulations to investigate the deposition, desorption, and re-adsorption of heterogeneous molecular films on a gold substrate in a representative space environment. The goal is to model molecular contamination processes relevant to spacecraft surfaces, such as those of quartz crystal microbalances (QCM) or sensitive optical components. Molecular dynamics enables atomistic-level resolution of these phenomena by numerically integrating Newton’s equations of motion to compute the time evolution of particles interacting under specified potential energy functions.^{6,7} The simulations were carried out using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS). We considered enhancing the simulations with the PLUMED 2 plugin to implement “metadynamics” for accelerated sampling of slow desorption processes⁸ although such results are left for a future work. The simulations discussed in this paper present the baseline obtained without any artificial acceleration techniques.

Molecular Dynamics is essentially a particle tracing algorithm in which the interaction between different atoms is governed through various force fields. A significant challenge in setting up an MD simulation is in the identification of the models to utilize. For example, the water molecule can be represented by considering from 3 to up to 6 “sites”, which correspond not only to the actual hydrogen and oxygen atoms, but to also virtual interaction points aiming to capture the electrical polarity and other properties of the molecule. Increasing the number of sites greatly increases the overall simulation cost due to the N^2 coupling between the interacting constituents. In our case, water-water interactions were modeled using the TIP4P/Ice force field, since it leads to accurate representation of water’s phase diagram at low temperatures and pressures,^{9,10} particularly for amorphous and solid forms of ice that are the most common forms of water-ice in space environments.

Water-gold interactions were modeled using a Morse potential,¹¹ which was found to be the best water-gold force field available based on detailed benchmarking studies against density functional theory (DFT) calculations of water adsorption on Au(111) and Au(001) surfaces. Unlike simpler Lennard-Jones potentials, the Morse potential accurately reproduced the DFT-calculated energy landscape, achieving excellent agreement across a range of molecular orientations and distances, with an R^2 value of 0.97. The Morse potential’s exponential terms for both attraction and repulsion allow independent tuning of equilibrium bond distance, well depth, and interaction stiffness, providing a physically meaningful fit to water-gold adsorption energetics. This potential correctly captured the preferred flat orientation of water molecules on gold and the balance of weak Au–H repulsion and Au–O attraction, both of which are essential to accurately model contaminant film behavior on metallic spacecraft surfaces. The general form of the Morse potential equation, where $V(r)$ is the potential energy between two atoms separated by distance r , D_e is the depth of the potential well (bond dissociation energy), r_e is the equilibrium bond length, and a is a parameter controlling the width of the potential well, is given below:

$$V(r) = D_e \left(1 - e^{-a(r-r_e)}\right)^2 \quad (1)$$

Hydrocarbon-hydrocarbon interactions were described using the OPLS-AA force field⁷, which was specifically parameterized to reproduce both gas-phase conformational energetics and condensed-phase thermodynamic properties of organic liquids. OPLS-AA accurately captures heats of vaporization, densities, and structural properties of hydrocarbons, making it well-suited for modeling molecular films of small organic contaminants. OPLS-AA is the most commonly used force field for hydrocarbons and many organic liquids in molecular dynamics. Hydrocarbons investigated in our simulations include decane ($C_{10}H_{22}$), toluene (C_7H_8), and methane (CH_4).

Nitrogen-nitrogen interactions were modeled with a simple Lennard-Jones “12-6” potential (the number indicate the powers of the repulsive and attractive terms), and remaining cross-species interactions were modeled using L-J potentials with Lorentz-Berthelot combining rules, except for water-gold interactions which utilized the previously described Morse potential. The Lennard-Jones potential is given by:⁶

$$V_{LJ}(r_{ij}) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^6 \right] \quad (2)$$

where r_{ij} is the distance between atoms i and j , ϵ_{ij} is the well depth, and σ_{ij} is the collision diameter.

The Lorentz-Berthelot combining rules define the cross-species interaction parameters as:⁶

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2} \quad (3)$$

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j} \quad (4)$$

where σ_i , σ_j , ϵ_i , and ϵ_j are the Lennard-Jones parameters for species i and j , respectively.

The gold substrate atoms were modeled using the Embedded Atom Method (EAM)¹³ for all gold-gold interactions. The EAM models metallic bonding by embedding each atom into the collective electron density formed by its neighbors, effectively capturing the influence of the surrounding electron cloud on cohesion, elastic properties, and defect energetics. The method accounts for many-body interactions by representing metallic bonds as a function of local electron density, which is beyond the capabilities of simpler L-J potentials. Due to the use of this force-field in this simulation, “metal” units were used instead of the standard “real” LAMMPS units, which required all of the force field parameters including bond coefficients, angle coefficients, dihedral coefficients, and potential well depth to be converted from real to metal units since they were often given in “real” units in their original literature.

In this simulation, “non-reactive” force-fields were chosen over more complex “reactive” force-fields such as ReaxFF.¹⁴ In short, reactive force fields are desired when chemical reactions (e.g. covalent bond breaking or formation) occur or dominate interactions in your simulated system, but aren’t necessary when such reactions don’t occur. Other than methane potentially reacting with the gold surface, none of the other molecule or atom pairs in the simulation are generally reactive with each other, and thus non-reactive force-fields were used. With

non-reactive force fields, there are several key advantages. First and foremost, they are often computationally much more efficient and thus significantly faster. Secondly, there are many more pre-existing non-reactive molecular force field models, particularly for water (such as the state-of-the-art TIP4P models), whereas for reactive force-fields, it might be hard to find force fields which are developed for all molecules of interest in your system. Creating our own molecular models was not feasible, as such an effort could easily be a Ph.D. dissertation-worthy topic on its own. As an example, water in particular is notoriously one of the most difficult models to model and simulate due to its array of unique properties, but there are a much larger number of non-reactive models which are able to simulate it under conditions which we're interested in here such as the low pressure low temperature phases of ice, whereas there's no widely validated reactive force field which can do the same.

All of the rigid molecular species (nitrogen and TIP4P water) in the simulation were made rigid via LAMMPS' `fix shake` command. The `fix shake` approach is generally far superior to the alternative `fix rigid` method for integrating rigid molecules in LAMMPS, since it allows a much larger timestep (up to 2 fs) and is much more computationally efficient. However, it also required several workarounds when implementing it in our code, particularly when depositing new rigid molecules (see discussion on molecular deposition below). Additionally, not only is `fix shake` able to run significantly faster than `fix rigid` due to its ability to use much larger timesteps, but it is also much more efficient in terms of parallel scalability, whereas `fix rigid` requires multiple collective communications to collect and update the rigid body information across all processors.

The simulation domain consisted of a gold substrate with periodic boundary conditions applied in the lateral directions to emulate an infinite surface. Domain dimensions were 73.44 x 36.72 x 150.0 Å (in the x , y , and z directions respectively), with the x and y dimensions matching the size of the gold brick upon which molecules would deposit and desorb from (with the height of the brick being 7.14 Å). This size was chosen both to balance relative computational feasibility and efficiency with a domain size that would reduce numerical artifacts as much as possible. The z -dimension is significantly larger in order to avoid any periodic effects in the z -dimension (this is also aided by the reflective and evaporative walls which I place above the gold substrate and deposition regions but well below the height of the simulation domain as well). We should also note that the x and y dimensions of the simulation domain and gold slab are exact multiples of gold's FCC lattice constant (4.065 Å) in order to ensure a realistic crystal structure and avoid periodic boundary artifacts in those directions.

The vertical (z) dimension also technically has a periodic boundary condition, however in reality, it is bounded at the top by a reflective barrier that elastically redirected any particles with upward initial velocities back toward the substrate (in the later desorption simulations, this reflective barrier was removed and replaced with an open boundary which dynamically removed molecules that crossed it utilizing the `fix evaporate` command). Above the reflective barrier, there is a large amount of empty vertical space before the actual top periodic boundary in order to prevent any periodic force-field effects in the surface-normal z direction. The bottom two layers of the gold substrate are anchored in place using the "`fix setforce 0.0 0.0 0.0`" command for those layers in order to provide a stable platform on which the remaining simulated molecules can interact. The minimum lateral domain dimensions were selected to exceed the long-range interaction cutoffs of the force fields employed, ensuring that particle-particle forces were accurately resolved. Gold lattice constants were used to define the x and y domain extents, resulting in minimum dimensions larger than gold's lattice constant of 4.065 Å, while the z -dimension was substantially larger to accommodate the deposition process.

A particle-particle particle-mesh (PPPM) solver was employed to compute long-range Coulombic interactions in the simulation.⁷ The PPPM method offers significant performance advantages over traditional Ewald summation for large systems which are less computationally efficient and often slower as a result. By mapping atomic charges onto a three-dimensional mesh and solving Poisson's equation via fast Fourier transforms (FFTs), PPPM ensures accurate electrostatics under periodic boundary conditions. When using electrostatics in LAMMPS, it's a best practice to select a domain size of at least 10–12 Å in each direction in order to provide sufficient resolution for the PPPM mesh and to prevent non-physical artifacts and spurious interactions between periodic images (however, our eventual domain size was much bigger, but noting this minimum size requirement is a small but very critical detail for anyone who wishes to make their own MD simulations in the future).

Both PPPM and Ewald summation methods provide accurate long-range Coulombic interactions under periodic boundary conditions. However, the PPPM method, specifically the `pppm/tip4p` style was employed with a

target relative RMS force error of 1.0×10^{-5} to ensure highly accurate treatment of long-range electrostatics, particularly for the off-site charges present in the TIP4P water model. This strict tolerance minimizes electrostatic artifacts that could affect water and general contamination layer behavior.

A Langevin thermostat was utilized to maintain the gold substrate at a constant temperature during deposition, and initially was used to heat it at a constant rate when simulating a TGA ramp before ultimately switching to thermostating all atoms with a Berendsen thermostat for the final TGA desorption simulations. Importantly, to enable effective energy transfer between the contaminant molecules and the gold surface, only the middle layers of gold were thermostatted during deposition: the bottom layers were fixed to anchor the slab as mentioned above, and the topmost gold layers were left unthermostatted so that they could interact freely and have realistic (non-dampened) energy transfer with the contaminants. While gold-gold interactions (between the non thermostatted top layer and the thermostatted middle layer) efficiently exchange energy due to their many-body metallic bonding and shared electron cloud, direct thermostating of the top gold layer would have dampened its natural fluctuations, limiting its ability to transfer thermal energy to the adsorbates at the interface. However, for desorption simulations, the Berendsen thermostat was applied to all atoms in the simulation since desorption simulations using both Langevin for the middle gold atoms and Berendsen for all atoms resulted in identical simulation results, except that the Berendsen thermostat ran much more quickly.

The simulation timestep was set to 1.54547 fs (or 0.00154547 ps, in the alternate “metal” units). Simulations were run for tens to hundreds of millions of time steps. Visualization data is saved to “.xyz” formatted files via the `dump` command and those are the files which are visualized in our visualization software of choice - `OVITO` or `Paraview`. The `write_data` command is also used at several key timesteps (although not every timestep) in order to save the simulation state to a restart file so that the final state (or chosen intermediate stages) of the simulation can then be reloaded and simulated in the future. Lastly, every time a new molecule is deposited in the simulation (or more importantly, deleted), the `reset_atoms mol all` command is used in order to properly re-number atoms and molecules between runs (unfortunately this is not handled automatically by LAMMPS due to a weird quirk, of which there are unfortunately many).

Molecular deposition was implemented by dynamically creating atoms using for loops that called LAMMPS’ `create_atoms` command. This approach was necessary due to the incompatibility of the native `fix deposit` command with the rigid body constraints enforced by `fix shake`. Unlike `fix rigid` or `fix rigid/small`, `fix shake` is unfortunately not compatible with LAMMPS’ native method for injecting new particles – which is `fix deposit` – so the `create_atoms` command is utilized in conjunction with for loops instead along with the additional steps described below. Prior to a new molecule being deposited into the simulation via the `fix_deposit` command, `fix shake` constraints are temporarily removed and then reapplied once the molecule is injected. The newly injected molecule was then assigned initial velocities drawn from Gaussian distributions corresponding to the desired simulation temperatures, and since LAMMPS unfortunately does not allow you to enforce directionality with randomly generated thermal velocities, a reflective boundary right above the molecular deposition region ensured net downward flux toward the substrate.

For the final deposition simulations, we ran scaling studies and two primary productions simulations - one *sandwich* case in which molecules of different types were deposited in layers, and a *heterogeneous* case in which molecules of all types were arriving randomly (with an increased weight assigned to water due to its relevant abundance in spacecraft outgassing). In the “sandwich” case, we simply built up an initial layer of several hundred water molecules on the gold substrate, then a layer of several hundred hydrocarbons, before then layering on a few hundred more water molecules. The sandwich deposition simulation results are shown in Figures 3 through 5 whereas the heterogeneous deposition results are shown in Figures 6-8, respectively. For the “scaling” studies however, three primary simulations were conducted and the final results seen in Figure 1. We also considered a sensitivity study to characterize the importance of the time interval between consecutive molecule generations. The reduced time intervals were desirable to the extreme length of the simulations. For example, our earlier effort with molecules being created only once every 50,000 time steps produced under 5 monolayers of deposited material after nearly two weeks of consecutive run time. In the production deposition cases, 1,000 timesteps were used between water depositions for the sandwich case for the first and third layer, whereas 4,000 timesteps was used between hydrocarbon depositions. For the final heterogeneous deposition case, 10,000 timesteps was used between every deposition. 1,000 and 5,000 timestep interval simulations were also ran for the heterogeneous case

but both simulations failed due to not being able to handle the smaller time interval, so the 10,000 timestep results were used for the heterogeneous case. For the sandwich case, the final results took nearly 1.5 weeks and several restarts to run (due to the two-day time limits imposed by the USC HPC cluster), so using timestep intervals much greater than 1,000 was infeasible. For the TGA cases, we ramped both the sandwich and heterogeneous cases up to 150K, 250K, 350K, and 450 K before then holding them there for several million timesteps beyond that. Those results are summarized in Figures 9 through 33 below with Molecule Counts over time for the 450 Kelvin TGAs given in Figure 15.

One important thing to note is that initial 450 Kelvin TGA simulations accidentally incorporated an incorrect version of Toluene which had unusually strong Lennard-Jones parameters for its hydrogens, however, we have included those results since these artificially sticky molecules could yield useful insights into the way that heavier or higher activation energy species behave in these scenarios. The TGA simulations which include the incorrect "Toluenes" are shown in Figures 16 through 23 below.

The equations of motion were integrated using the Velocity-Verlet algorithm,^{6,7} which is widely used in MD simulations for its stability and time-reversibility. The position $\mathbf{r}_i$ and velocity $\mathbf{v}_i$ of each atom i were updated according to:

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\Delta t + \frac{1}{2} \frac{\mathbf{F}_i(t)}{m_i} \Delta t^2 \quad (5)$$

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \frac{1}{2} \left[\frac{\mathbf{F}_i(t)}{m_i} + \frac{\mathbf{F}_i(t + \Delta t)}{m_i} \right] \Delta t \quad (6)$$

where $\mathbf{F}_i$ is the force on atom i , m_i is its mass, and Δt is the integration timestep. Timestep sizes were selected to ensure numerical stability, with values typically in the sub-femtosecond to femtosecond range depending on the molecular species and force fields involved.

Due to the intrinsic time scale limitations of molecular dynamics, where even with high-performance computing only nanoseconds of physical time can be simulated, metadynamics can be employed to enhance sampling of rare desorption events at low temperatures. We spent significant time implementing the PLUMED 2 LAMMPS plugin to implement well-tempered metadynamics into our simulations, with the coordination number selected as the collective variable to encourage the solid-to-gas phase transition (sublimation); however as mentioned above we did not ultimately utilize these results in this paper so that we could have a baseline of non-enhanced TGA data to compare to and validate against for future studies which ultimately do implement PLUMED. This approach would have allowed desorption at cryogenic temperatures, such as 150-300 K, to be simulated on accessible timescales while preserving the physics, which would otherwise require unfeasible wall times (perhaps even thousands or millions of years of real time at very low temperatures) without metadynamics or enhanced sampling using LAMMPS and current computational resources alone.

Simulations were primarily performed on the high-performance computing clusters at the USC Center for Advanced Research Computing (CARC). Custom LAMMPS builds with PLUMED integration were optimized for the specific simulation configurations. Job scheduling was managed through SLURM, with typical runs utilizing 32 CPU cores on a single node to balance parallel efficiency with inter-core/inter-socket communication constraints. Post-processing and visualization were conducted using the Open Visualization Tool (OVITO),¹⁵ which is one of the two primary visualization tools mentioned earlier in the paper. This data provided input for the development of reduced-order contaminant transport models designed to improve upon traditional sticking-coefficient-based approaches. Lastly, the exact force-field parameters for all of the molecules which were used, visualizations of the molecules, and additional information are included in Tables 1 through 4. Table 1 also includes visualizations of the molecules and the force-field parameters for each molecular interaction between like molecules. Table 2 includes all cross-molecule interactions, and Tables 3 and 4 include the bond, angle, and dihedral coefficients used in the simulation respectively. LAMMPS Input and Molecule Files can be provided upon request.

Table 1: Individual molecules used in the simulations, molecular visualizations, their force fields, and atomic force-field parameters. Hydrocarbons all use the same OPLS-AA parameters.

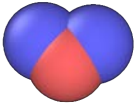
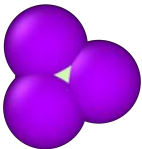
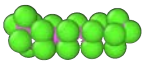
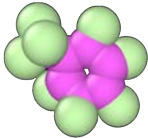
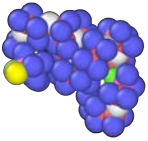
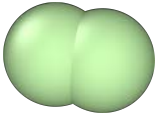
Molecule	Visualization	Model / Force Field	Force Field Parameters
Water (H_2O)		TIP4P/Ice	O – O: $\epsilon = 0.00914$ eV, $\sigma = 3.1668$ Å H – H: $\epsilon = 0.0$ eV, $\sigma = 1.0$ Å O – H: $\epsilon = 0.0$ eV, $\sigma = 1.0$ Å
Methane (CH_4)		OPLS-AA	C – C: $\epsilon = 0.00286$ eV, $\sigma = 3.5$ Å C – H: $\epsilon = 0.0$ eV, $\sigma = 0.0$ Å H – H: $\epsilon = 0.00130$ eV, $\sigma = 2.5$ Å
Decane ($\text{C}_{10}\text{H}_{22}$)		OPLS-AA	Same params as above
Toluene (C_7H_8)		OPLS-AA	Same params as above
PEG600		CHARMM36	O – O: $\epsilon = 0.01096$ eV, $\sigma = 2.85$ Å C – C: $\epsilon = 0.00288$ eV, $\sigma = 3.58$ Å O – C: $\epsilon = 0.00561$ eV, $\sigma = 3.19$ Å O – H: $\epsilon = 0.00367$ eV, $\sigma = 2.60$ Å C – H: $\epsilon = 0.00188$ eV, $\sigma = 2.92$ Å H – H: $\epsilon = 0.00123$ eV, $\sigma = 2.37$ Å
Nitrogen (N_2)		LJ 12-6	N – N: $\epsilon = 0.00615$ eV, $\sigma = 3.80$ Å

Table 2: Lennard-Jones and Morse potential cross-parameters used in the simulations.

Species Pair	Atomic Pair	ϵ_{ij} (eV)	σ_{ij} (Å)
Water – Hydrocarbons	O – C	0.00512	3.3334
Water – Hydrocarbons	H – C	0.0	2.25
Water – Hydrocarbons	O – H _{hydro}	0.00345	2.8334
Water – Hydrocarbons	H – H _{hydro}	0.0	1.75
Nitrogen – Hydrocarbons	N – C	0.00420	3.649
Nitrogen – Hydrocarbons	N – H _{hydro}	0.00283	3.149
PEG – Hydrocarbons	O – C	0.01079	2.83
PEG – Hydrocarbons	C – C	0.00553	3.17
PEG – Hydrocarbons	O – H _{hydro}	0.00361	2.58
PEG – Hydrocarbons	C – H _{hydro}	0.00188	2.92
Nitrogen – Water	N – O	0.00750	3.4824
Nitrogen – Water	N – H	0.0	2.399
PEG – Water	O – O	0.00675	3.03
PEG – Water	O – H	0.00288	2.65
PEG – Water	C – O	0.00440	3.26
PEG – Water	C – H	0.00188	2.92
Water – Gold	O – Au (Morse)	$D_e = 0.019278$, $r_e = 0.905$, $\alpha = 4.2$	
Water – Gold	H – Au (Morse)	$D_e = 0.00082914$, $r_e = 1.41$, $\alpha = 4.14$	

Table 3: Bond coefficients used in the simulations.

Type	Molecule/Bond	Force Constant (eV)	Equilibrium Length (Å)
1	OH (TIP4P/Ice)	0	0.9572
2	N $\equiv$ N (rigid)	0	1.1
3	C-H (Methane)	15.146	1.094
4	PEG	48.103	1.12
5	PEG	17.337	1.43
6	PEG	13.002	1.52
7	PEG	17.347	1.53
8	PEG	13.474	1.41
9	PEG	7.7732	0.971
10	Toluene C-H	15.146	1.09
11	Toluene C-C	17.341	1.39
12	Toluene C-C	17.348	1.4
13	Toluene	13.002	1.52
14	Toluene	0	0.0
15	Decane	15.146	1.09
16	Decane	17.347	1.53
17	Decane	15.174	1.1

Table 4: Equilibrium angle values and dihedral coefficients used in the simulations.

Type	Parameter	Force Constant (eV)	Equilibrium Value
1	Angle (Rigid Water)	0	104.52°
2	Angle (Methane)	2.0590	109.5°
3	Angle (PEG)	2.0682	109.5°
4	Angle (PEG)	2.3905	109.5°
5	Angle (PEG)	2.0656	109.6°
6	Angle (PEG)	2.3900	111.0°
7	Angle (PEG)	2.3847	110.3°
8	Angle (PEG)	7.8255	109.0°
9	Angle (PEG)	1.3084	110.0°
10	Angle (Toluene)	2.1756	108.0°
11	Angle (Toluene)	1.9595	120.0°
12	Angle (Toluene)	2.1733	120.0°
13	Angle (Toluene)	2.3860	111.4°
14	Angle (Decane)	2.0656	109.6°
15	Angle (Decane)	2.3900	111.0°
16	Angle (Decane)	2.2756	107.57°
17	Angle (Decane)	2.3861	106.75°
18	Angle (Decane)	2.0599	108.53°
19	Angle (Decane)	2.0590	109.5°
1	Dihedral (PEG)	0.013059	1 3
2	Dihedral (PEG)	0.061356	1 3
3	Dihedral (PEG)	0.010364	-1 3
4	Dihedral (PEG)	0.0	1 1
5	Dihedral (Toluene)	0.43323	-1 2
6	Dihedral (Toluene)	0.010364	1 6
7	Dihedral (Toluene)	0.0	1 1
8	Dihedral (Decane)	0.061356	1 3
9	Dihedral (Decane)	0.0	1 1

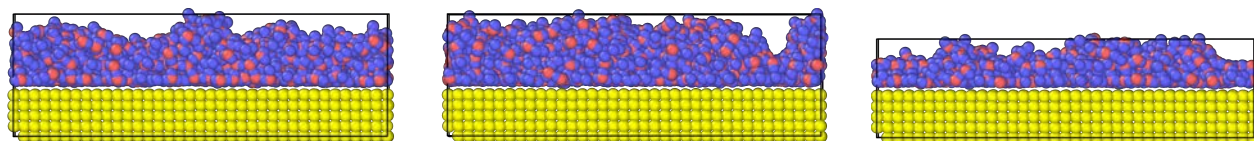
3. RESULTS

3.1 Sensitivity Study

For the two primary cases tested - the water-hydrocarbon-water “sandwich” and the heterogeneous cases - deposition ended up taking several days to simulate, even with the reduced inter-insertion interval. Initially, the plan was to run the simulation for 50,000 timesteps between depositions in order to preserve the “vacuum-like” conditions and give the molecules enough time to deposit and settle before the next one is generated, but this ultimately proved to be far too slow of a speed if we wanted to simulate our full deposition simulations within reasonable real-world timeframes - with these simulations taking several weeks or more to even reach the halfway point when using 50,000 timesteps or more for each deposition. Thus, as mentioned, we undertook several scaling studies with reduced timesteps of 1,000, 5,000, and 10,000 timesteps, respectively, and we analyzed the performance of those in order to evaluate the optimal timestep amount that we could use to reduce total simulation time as much as possible while having the smallest possible effect on the actual simulation results. In order to test this, we performed sensitivity studies on simulations where several hundred water molecules deposited onto the gold substrate at intervals of 1000, 5000, and 10000 timesteps between depositions, respectively.

Final results at the end of each simulation are visualized and compared in Figure 1. In the simulations, there is qualitatively a noticeable difference between the final configuration in the 1000 timestep case compared to the 5000 and 10000 timestep cases, whereas the 5000 and 10000 timestep cases are more or less similar visually. The 1000 timestep case is slightly “flatter” than the latter two cases and when visualizing the XYZ file in OVITO, the molecules are clearly “shaking more” until deposition stops, likely since molecules are being injected too frequently for the surface thermal oscillations to damp out and equilibrate. Furthermore, when frequent collisions between depositing molecules occur as it does relatively often in the 1000 timestep case, the area above the substrate is technically no longer a vacuum but instead is a high pressure gaseous environment, which may have an effect on the final surface configuration in addition to the reduced time for thermal equilibrium mentioned earlier. Collisions in the 5000 and 10000 timestep cases, on the other hand, were rare based on a visual observation of the time-dependent results. Ultimately, the final heterogeneous simulations used 10000 timestep intervals between depositions whereas the final “sandwich” simulation utilized 1000 timestep intervals for the water depositions and 4000 timestep depositions for the hydrocarbons due to real-world runtime constraints.

Heterogeneous simulations were also run at 1000 and 5000 timestep intervals but failed due to a solver crash. The 10000 timestep version didn’t generate this error, and hence that option was selected for the final heterogeneous simulations. The accelerated 1000 and 4000 timestep configuration was found to be necessary in the sandwich configuration for the water and hydrocarbon deposition intervals, respectively, due to the necessity to perform frequent restarts because of the HPC time limits. This might’ve resulted in slightly less accurate deposition results for the sandwich case, and could be something that’s investigated further in the future.



(a) 1,000 timestep deposition interval (b) 5,000 timestep deposition interval (c) 10k timestep deposition interval
Figure 1: Final configurations of the water-only sensitivity study cases.

3.2 Deposition

Figure 2 shows the initial state for the sandwich and heterogeneous configurations with just the gold slab and a single molecule which was added at a random location in the space above the gold substrate. Figure 3 next plots simulation results from the sandwich configuration, just after the initial water layer deposition sequence is completed. Deposition of hydrocarbons continues next, as visualized in Figure 4. The final configuration is shown in Figure 5. Some interesting observations can be made from these visuals.

The sandwich case resulted in clean separation between the first water layer and the intermediate hydrocarbon layer, with virtually none of the hydrocarbons successfully penetrating through the (relatively thin) water layer

which deposited before it. Furthermore, with the intermediate hydrocarbon layer - which again consists of Decane, Nitrogen (technically not a hydrocarbon but still included), Methane, and Toluene - the molecules seem to be relatively evenly distributed, and for the most part, "lighter" species aren't significantly displaced by the "heavier" species, at least at the substrate temperature of 73 Kelvin and timestep-interval of 4,000 timesteps tested here. The top layer of waters, however, deviates from that behavior and instead of forming a completely separate, "clean" film on top of the hydrocarbon layer, the water molecules do sink into the hydrocarbons a decent amount and also coalesce together into clusters instead of forming a film on top. This could be due to the extremely small 1,000 timestep interval which was necessary to simulate a system of this size, or could be the actual physics, but it's something which could be investigated in future studies.

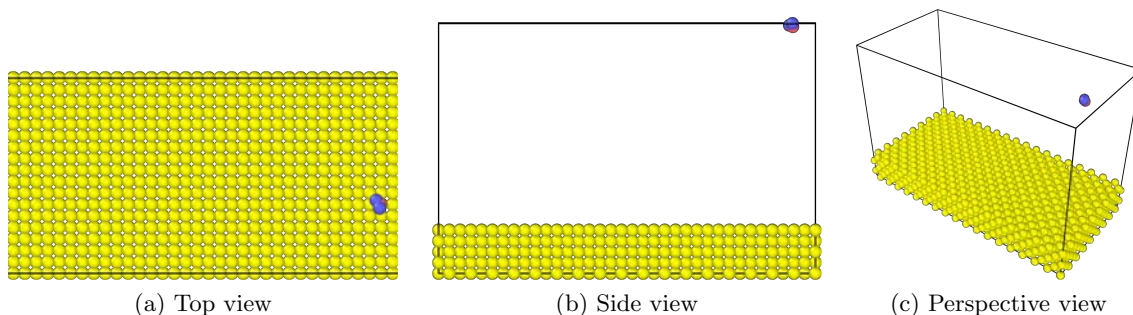


Figure 2: Initial configurations for both sandwich and heterogeneous deposition cases.

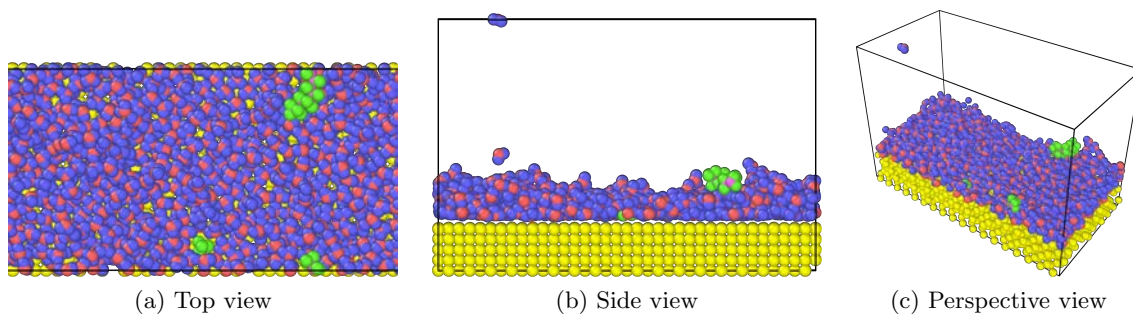


Figure 3: Sandwich case after the first water layer formed

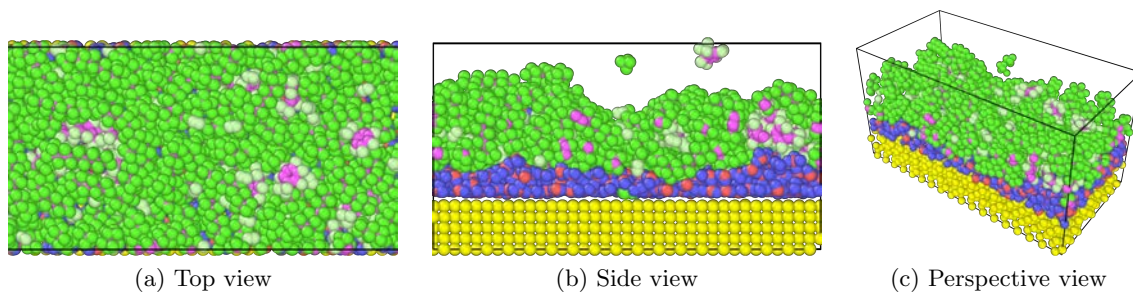


Figure 4: Sandwich case: Configuration after hydrocarbon layer formation

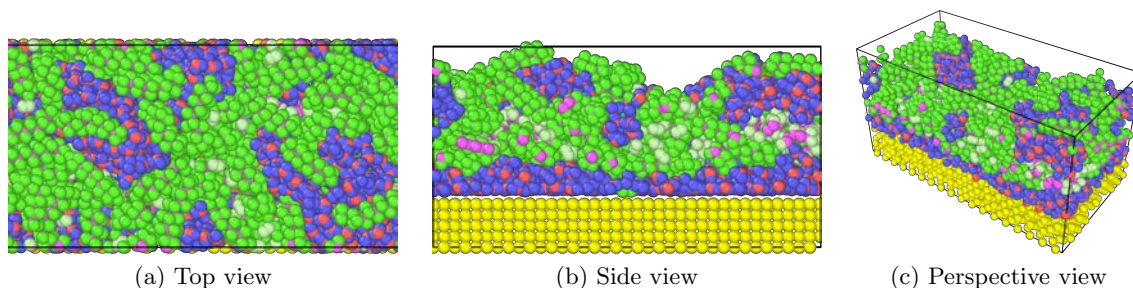


Figure 5: Final configuration of the sandwich deposition case, after the third (water) layer formed

Figures 6 and 7 visualize the deposition behavior for the heterogeneous case. The first figure corresponds to 2 million time steps, while the second is at the end of the simulation. Due to utilizing the longer 10,000 timestep deposition-interval, the final configuration of the heterogeneous deposition simulation included less molecules, but, on the flip side, potentially reflects more accurate deposition physics. Figure 8 is the result of restarting and running the Heterogeneous simulation for two additional days (with the new total time being 2x the original simulation time), although due to the decrease in computational efficiency as molecules are injected, much less than 2x the original amount of molecules are deposited. While further study will be needed, the water did not appear to displace any of the already deposited hydrocarbons, at least not the heavier ones such as decane or toluene, at the deposition-interval (10,000 timesteps between depositions) and deposition temperatures tested. In the future, it could be worth re-running the top layer of the sandwich simulations at a larger timestep-interval between timesteps to see if similar behaviour would occur and result in a more "uniform" layer rather than the current result where we see the water forming "islands" in the sandwich hydrocarbon layer with those "islands" also "sinking" into it quite a bit.

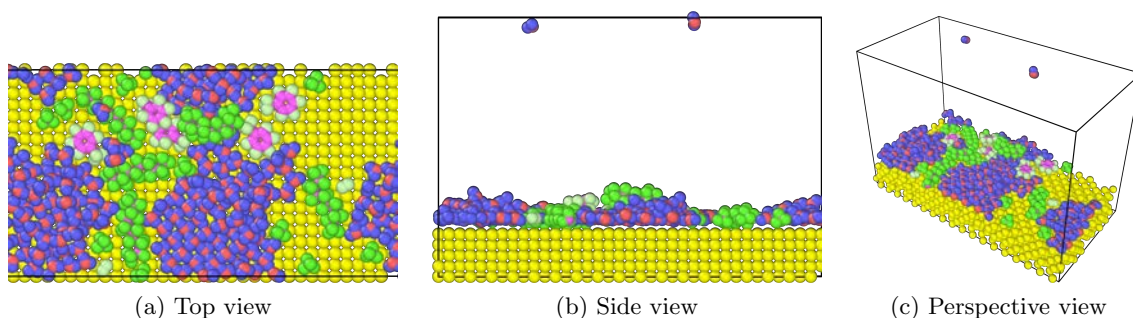


Figure 6: Heterogeneous deposition case at timestep 2 million (first monolayer partially formed).

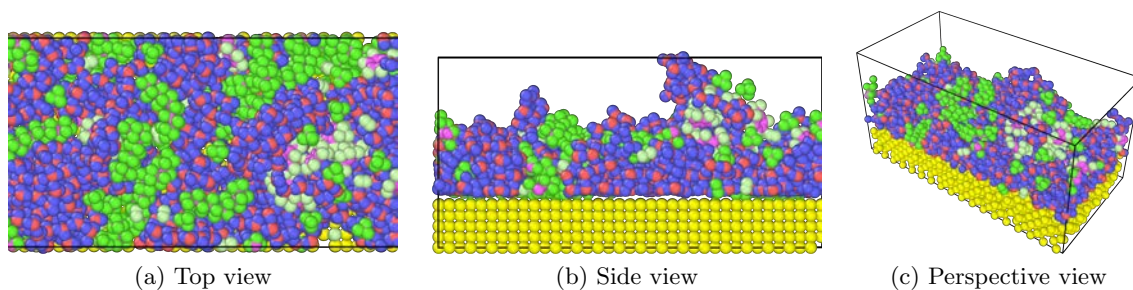


Figure 7: Final configuration of the heterogeneous deposition case.

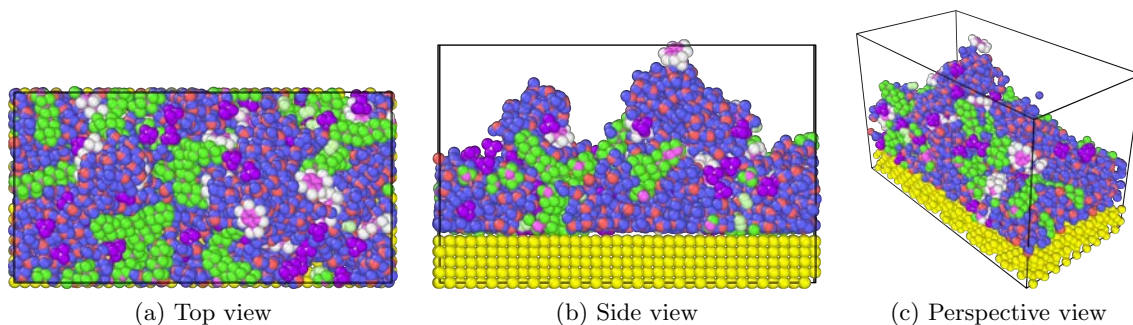


Figure 8: Final configuration of the heterogeneous deposition case with 2x the original simulation time.

3.3 Desorption

Results from these two deposition cases were used as the initial condition for a sequence of desorption cases. Figures 12 and 16 plot the initial configuration for our first batch of simulations with the sandwich configuration with the substrate temperature set to 450 K. By timesteps 324,000 and 576,000, respectively, we can observe relaxation of the adsorbed material, which is visible in the increased thickness of the surface layer. Water can also be seen to sink to the gold substrate. This is shown in Figures 13 and 17. Figures 18 and 14 plot the simulation after the decane layers have desorbed. For the incorrect-toluene simulations, the water layer is completely gone after 1.1 million time steps, leaving behind only toluene as shown in Figure 19, and for the sandwich case with the correct toluenes, no molecules are left behind. However, as noted earlier, the toluene model used for these initial simulation contained an error in its potential model, leading to the excessive sticky behavior.

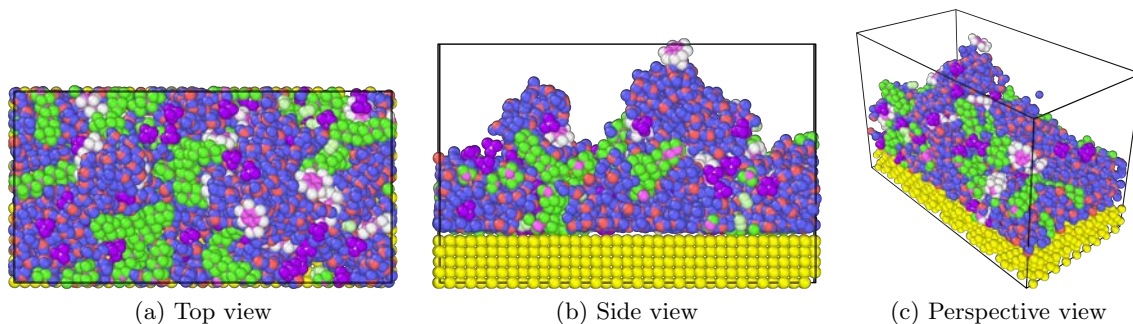


Figure 9: 450K heterogeneous simulation at timestep 0 (initial configuration).

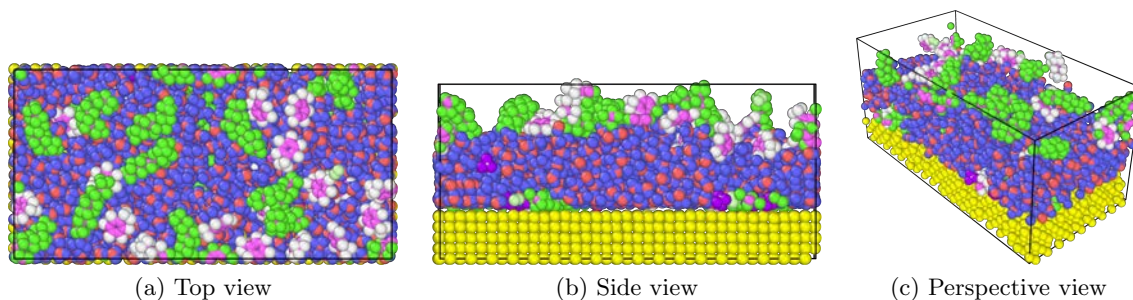


Figure 10: 450K heterogeneous simulation at timestep 324,000 (after dynamic restructuring).

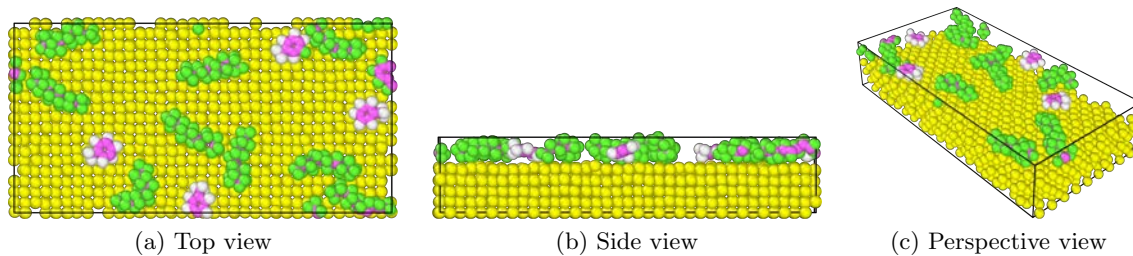


Figure 11: Final configuration of 450K heterogeneous case at timestep 2.5 Million, showing residual contaminants.

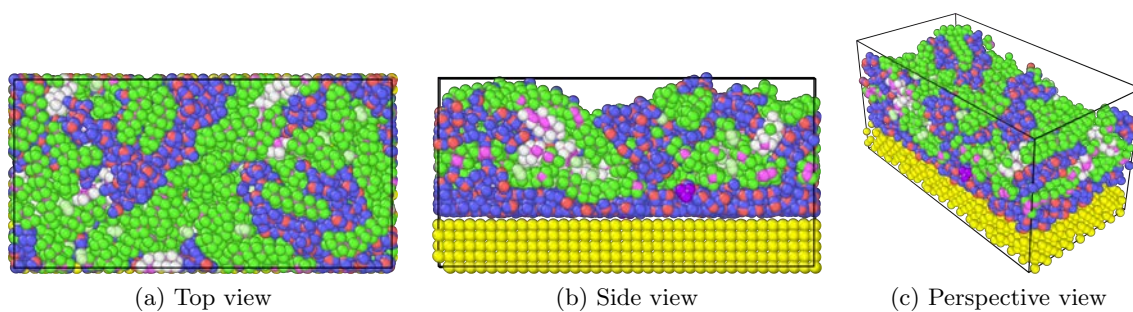


Figure 12: 450K sandwich configuration at timestep 0 (initial configuration).

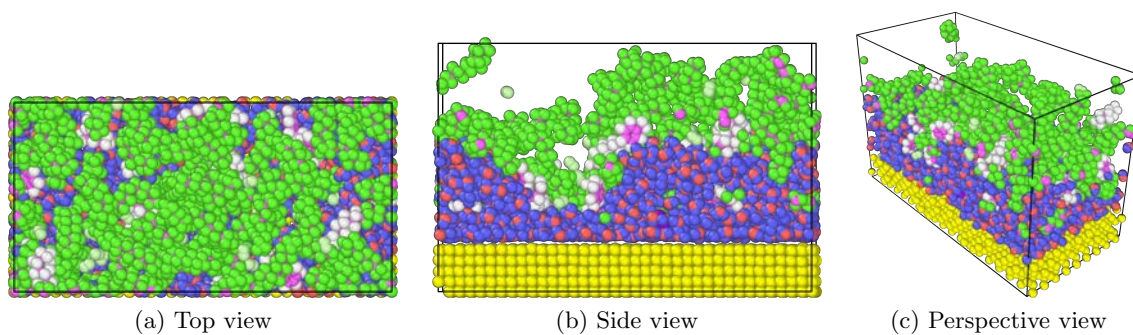


Figure 13: 450K sandwich configuration at timestep 300,000.

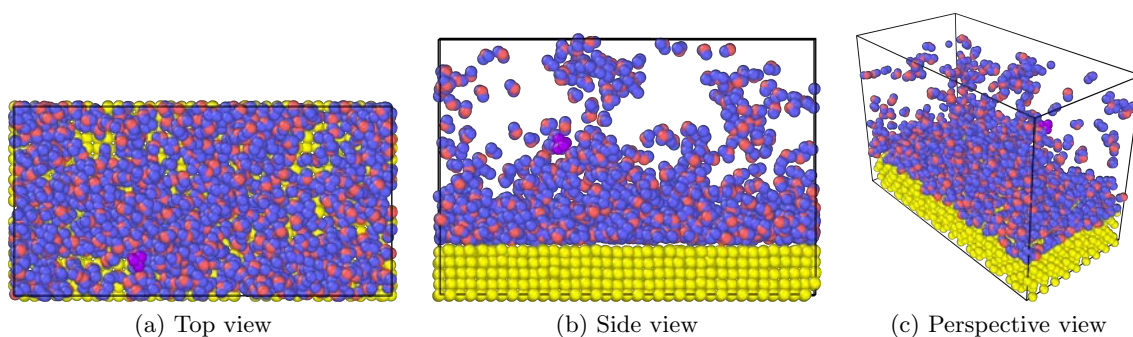


Figure 14: 450K sandwich case at timestep 386,000 (hydrocarbon layer desorbed, water layer beginning to desorb).

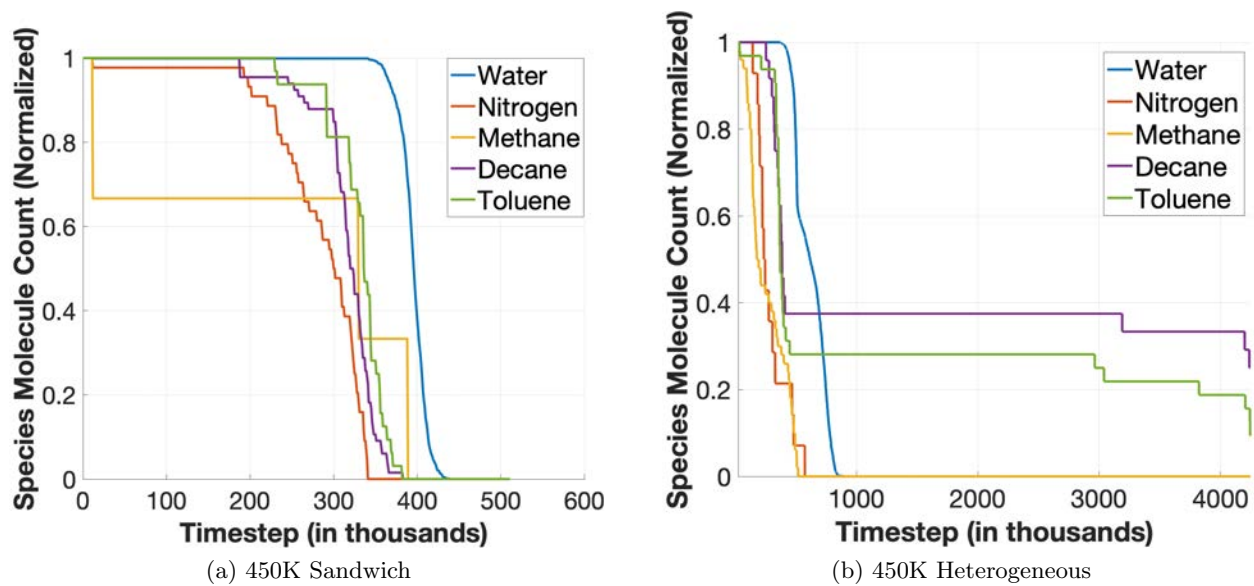


Figure 15: Normalized molecular counts vs time for the two desorption simulations at $T_{sub} = 450$ K.

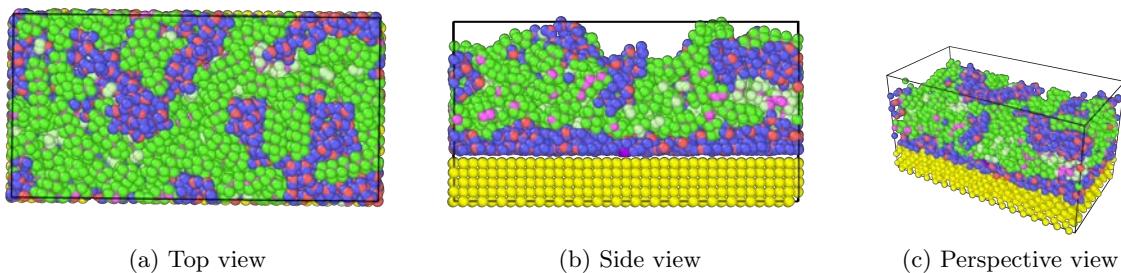


Figure 16: 450K sandwich TGA (bad toluenes) at timestep 0.

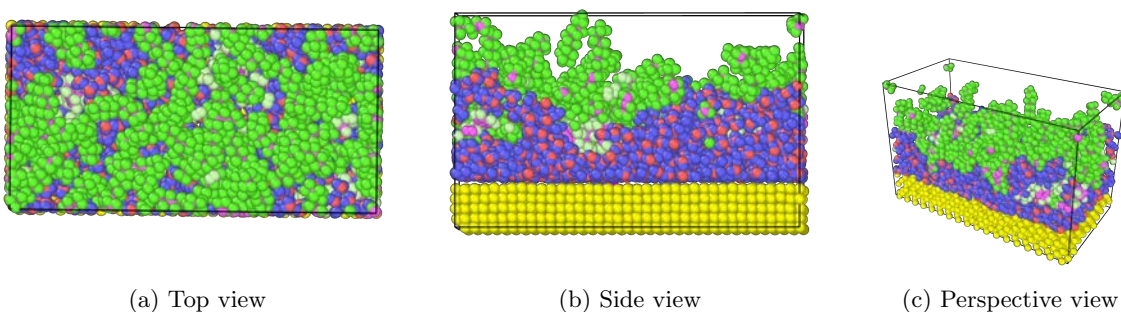


Figure 17: 450K sandwich simulation (bad toluenes) at ts 576,000 after dynamic layer rearrangement.

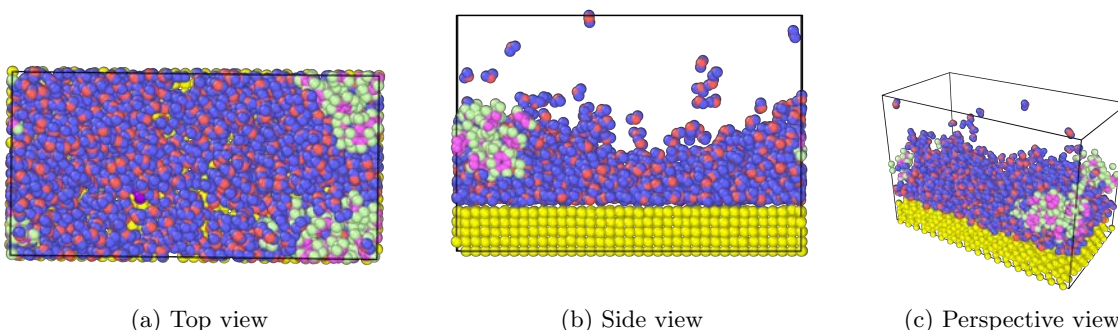


Figure 18: 450K sandwich simulation (bad toluenes) at ts 702,000 after decane layer desorption.

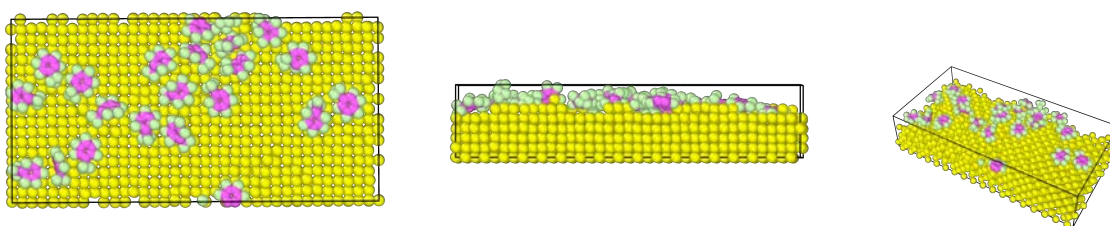


Figure 19: 450K Sandwich TGA simulation (incorrect toluenes) at timestep 1.1 Million with residual "toluenes" remaining behind. These simulations utilized incorrect potential model for toluene, leading to this extra sticky behavior.

Comparable results for the heterogeneous configuration are shown in Figures 20 through 23 for the incorrect toluenes and Figures 9 through 11 for the cases with the correct toluenes. The pictures in the series show that just as before, desorption begins with a rearrangement of the layers. Next, molecules start leaving the outermost monolayer. Once water molecules completely desorb, we are left with some residual decanes and also toluenes remaining on the gold substrate. It's important to note that those that remain behind were either initially deposited on the gold substrate or very close to it during dynamical restructuring (the only exception being the artificial/fake toluenes "sinking" down in that case just as they do in the sandwich simulations).

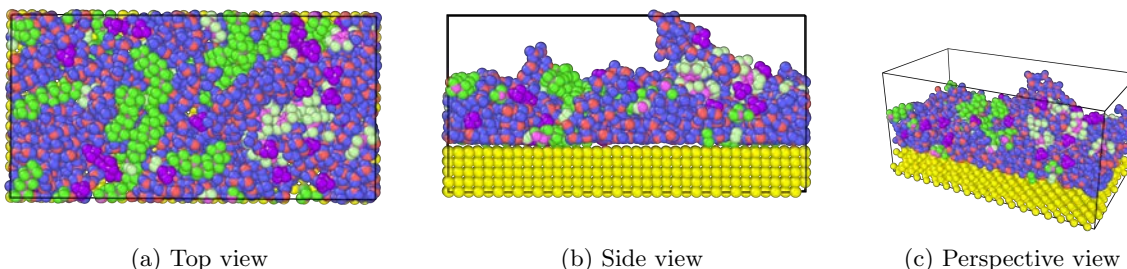


Figure 20: 450K Heterogeneous TGA simulation (bad toluenes) at timestep 0.

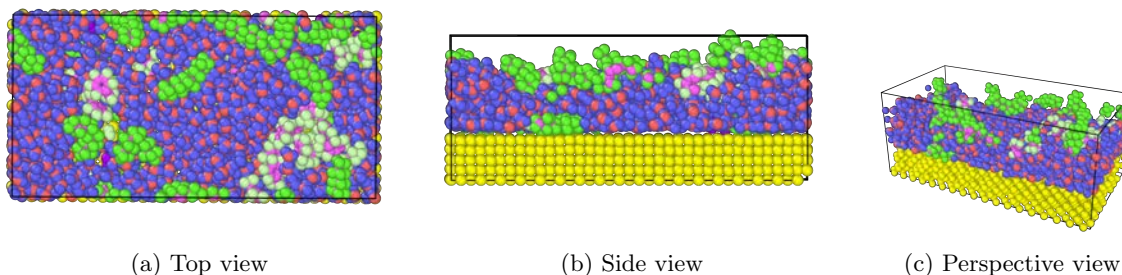


Figure 21: 450K Heterogeneous TGA (bad tolunes) at timestep 566,000 showing the rearranged layers.

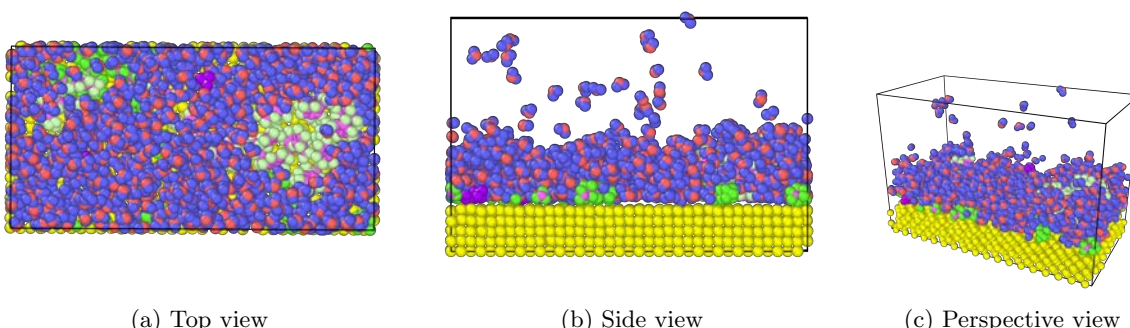


Figure 22: 450K Heterogeneous TGA (incorrect tolunes) at timestep 966,000 after the previous hydrocarbon layer has desorbed.

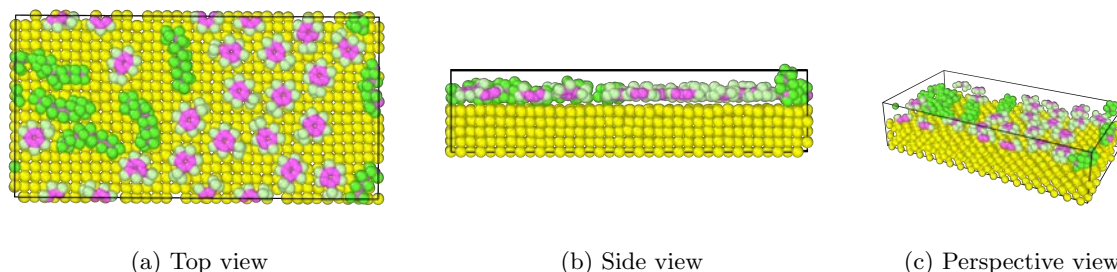


Figure 23: 450K Heterogeneous TGA (incorrect tolunes) at timestep 3.2 Million showing the final desorption configuration with all of the original "toluenes" and even some decanes remaining behind.

3.4 Surface Temperature

We next considered desorption behavior at additional, more realistic temperatures. The reason for selecting the initial hot 450 K configuration was so that the simulation could run in a reasonable time. Unfortunately as the substrate temperature decreases, the run time needed to establish some sort of response grows exponentially, at least according to the widely utilized Arrhenius model. As part of our future work, we plan to investigate the use of "metadynamics" to accelerate these slow processes at lower temperatures.

The initial state for these simulations is the same as shown previously for the 450 K configuration. Figure 24 then plots the sandwich configuration after 372,000 time steps at 350 Kelvin. Again we can see an increase of the deposited layer. At time step 534,000 we observe hydrocarbons such as decane and toluene desorbing into the

gas phase. Interestingly, we do not observe much water desorption, at least initially. Instead, water molecules seem to rearrange into a film in contact with the gold substrate. However, by time step 711,000, Figure 26, all water is gone as well, leaving behind a clean substrate.

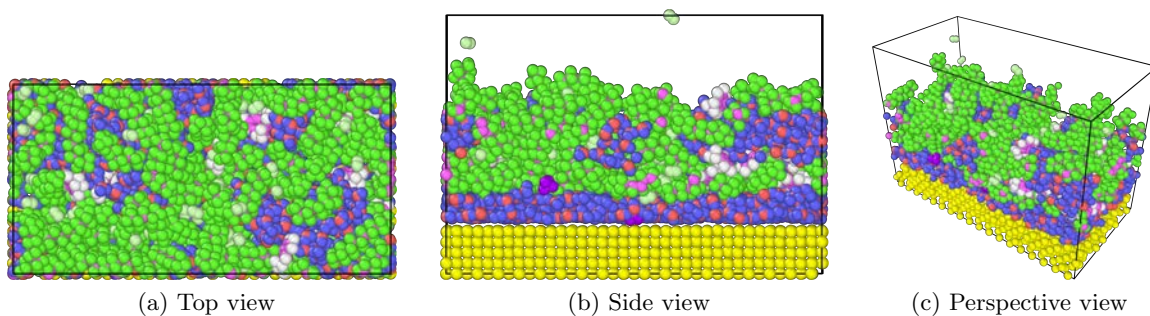


Figure 24: Sandwich TGA simulation for $T_{sub} = 350$ K at $ts = 372,000$.

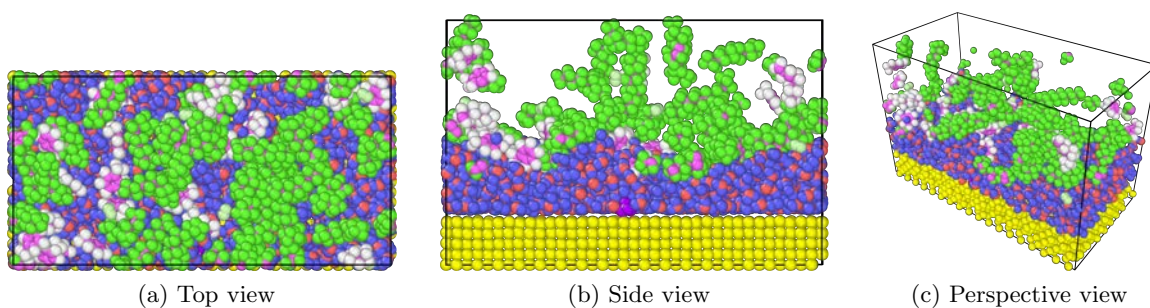


Figure 25: Sandwich TGA simulation for $T_{sub} = 350$ K at $ts = 534,000$.

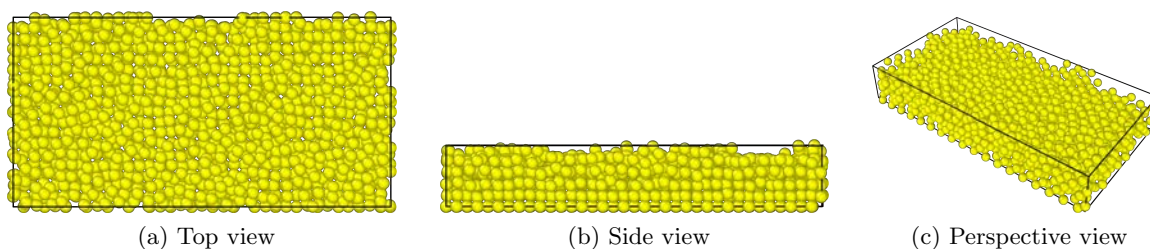


Figure 26: Sandwich TGA simulation for $T_{sub} = 350$ K at $ts = 711,000$.

The corresponding simulation for the heterogeneous configuration is plotted next. In Figure 27 we see methane molecules desorbing at time step 138,000. Separate water clusters seen in Figure 28 can be seen to have merged by time step 498,000 in Figure 28, leading to a configuration somewhat similar to the prior sandwich setup, although some decane and other hydrocarbon molecules can be seen to be adhered to the gold. This simulation still shows water adhered to the surface at the conclusion (see Figure 29) since while it would likely desorb, needed more runtime before desorbing at this temperature. Similar final results are given for 150 K and 250 K cases in Figures 30 through 33, respectively, although these would likely need to implement metadynamics in order to simulate on real-world timescales and as mentioned, will likely be the investigation of future work.

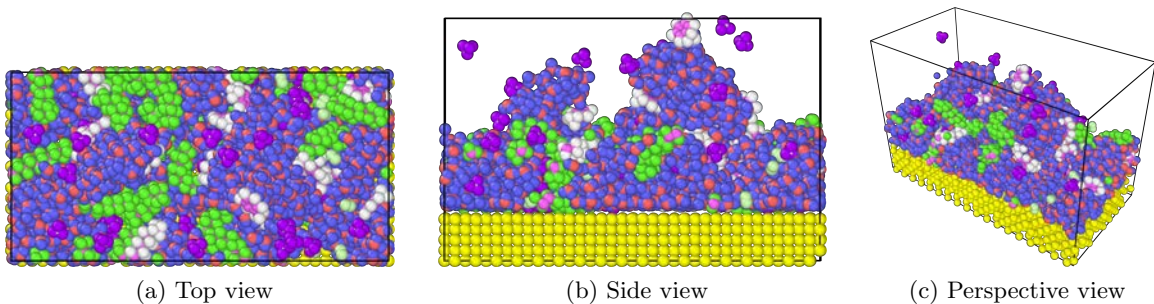


Figure 27: Heterogeneous desorption simulation for $T_{sub} = 350$ K at $ts = 138,000$.

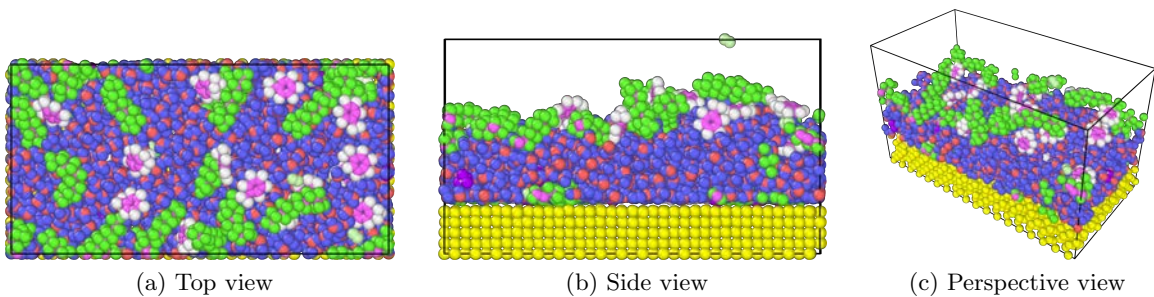


Figure 28: Heterogeneous desorption simulation for $T_{sub} = 350$ K at $ts = 498,000$.

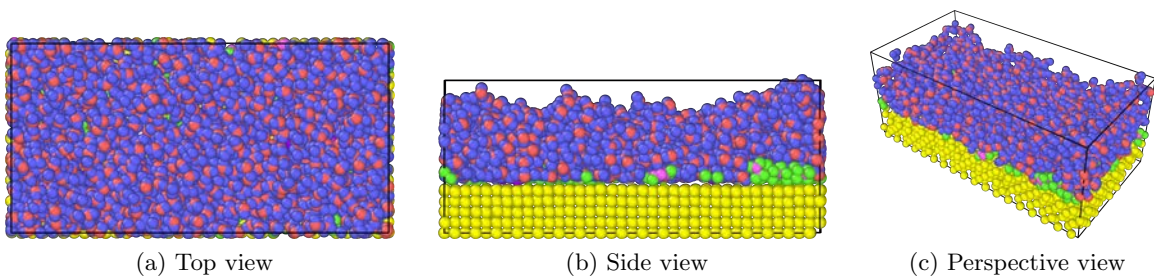


Figure 29: Heterogeneous desorption simulation for $T_{sub} = 350$ K at $ts = 1,152,000$

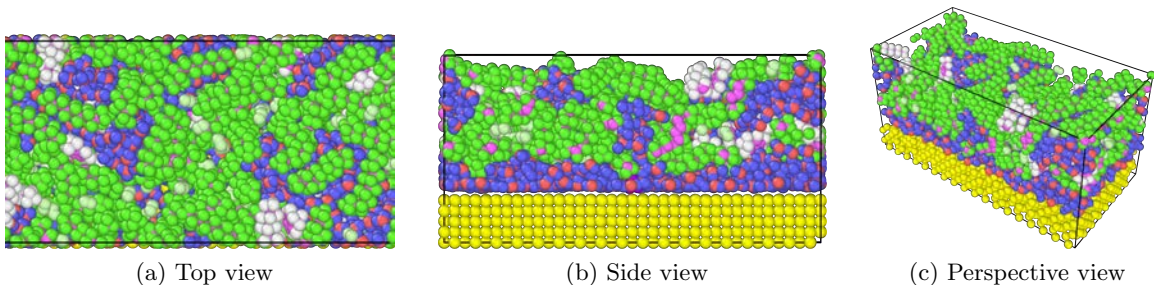


Figure 30: Final Sandwich TGA configuration for $T_{sub} = 150$ K at $ts = 1.2$ Million.

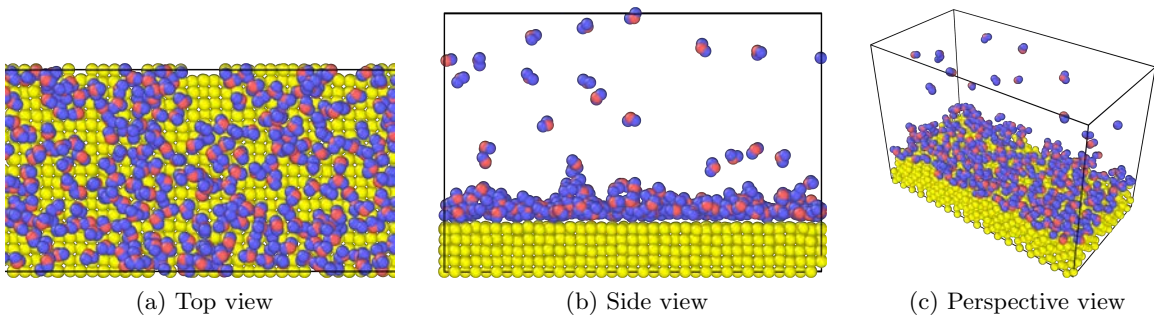


Figure 31: Final Sandwich TGA configuration for $T_{sub} = 250$ K at $ts = 1.07$ Million.

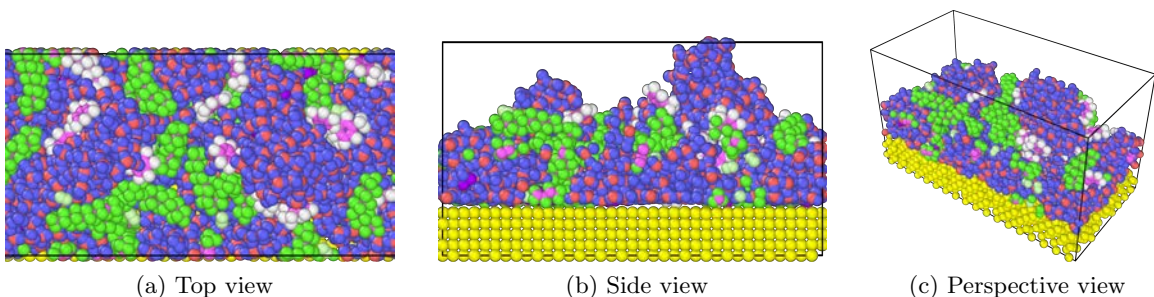


Figure 32: Final Heterogeneous TGA configuration for $T_{sub} = 150$ K at $ts = 1.045$ Million.

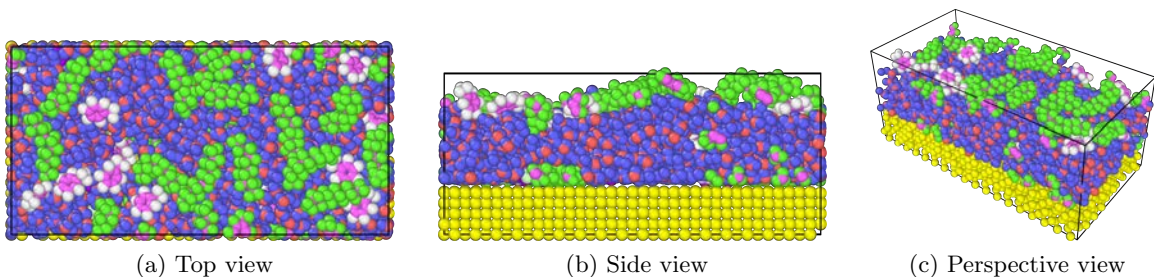


Figure 33: Final Heterogeneous TGA configuration for $T_{sub} = 250$ K at $ts = 1.053$ Million.

4. DISCUSSION

For the two primary cases – heterogeneous and sandwich – we largely observe the same behaviors. First, lighter species such as nitrogen and methane desorb from the substrate the earliest by far (unless if trapped deep into the substrate or if adsorbed to the gold). Then, the Decanes and Toluenes are the next species to desorb, and then finally, the water molecules ultimately desorb. In the cases where the "heavy" Toluenes were used, the Toluenes aren't carried away by the water and instead all "sink," remain behind, and stick to the gold surface. This could suggest that stronger activation energy species tend to generally stay behind as "lighter" species desorb sooner, for the most part. The Toluenes with the correct parameters however, desorb at roughly the same time as the Decanes more or less as seen in the visuals and in Figures 15a and 15b.

The only difference in behavior between the heterogeneous and sandwich simulations is that a few Decane molecules remain behind by the end of the heterogeneous simulations as a part of the incomplete monolayer of molecules still attached to the gold (and some toluenes also). Essentially, this shows that decanes or toluenes which were originally deposited on gold is more likely to remain than the decanes and toluenes that deposited

on water. In other words, the activation energies were different for different substrates, which is one of the key insights of these numerical simulations. None of the hydrocarbons remained behind in the sandwich simulations likely because of the water layer in between. As mentioned earlier though, in the simulations with the incorrect toluene molecules, the "heavy" toluenes generally tend to "sink" to the bottom even as all other molecules desorb, again suggesting that "heavier" species with much stronger binding energies would tend to remain behind as other contaminants fly away.

On the other hand, intermixing of lighter and heavier molecular species does not appear to completely "trap" lighter species. Instead, the lighter species rise to the top and tend to desorb first while the system also rearranges itself such that "heavier" species "sink" to the bottom and tend to desorb last. Thus, while temperature is still a significant factor in desorption, this all suggests that the activation energy based on the source-target material pair is also a very significant factor. "Trapping" does not seem to be a very significant effect, however, as much as source-target material does, with the one possible exception being species of completely different binding energies which are so different that the much "lighter" species might be trapped until the "heavier" species begins to rearrange/desorb (e.g. single nitrogens or methanes trapped deep inside a water crystalline structure for example...).

We also ran several simulations similar to that in Figure 1 for all molecules used in the simulation and then performed TGAs of all of those single-molecule configurations. The results further validated what we essentially saw in the heterogeneous and sandwich TGA simulations, which is that different target-source material pairs will have different activation energies, with that being demonstrated in all of these cases with a single monolayer (for all cases) remaining on the gold surface well after all of the other molecules had desorbed with the only exception being water, which has different properties and interactions with both gold and itself compared to the rest of the contaminants. Furthermore, as already mentioned, the contaminant molecules all desorbed in the same relative order as they did - for the most part - in the mixture TGA simulations as seen in Figures 15a and 15b, with only a few exceptions being for lighter molecules that were either trapped by the water and/or remaining behind due to the stronger attraction to the gold substrate due to the same effect described earlier in this paragraph.

5. CONCLUSION

This study demonstrates that molecular dynamics simulations can reveal complex interactions in heterogeneous contaminant films and reveals several key insights, particularly regarding both the importance of the source-target material pairings on desorption as well as the phenomena of dynamic restructuring in desorption. By comparing both layered ("sandwich") and randomized ("heterogeneous") configurations, we observed that molecules such as hydrocarbons desorb earlier when deposited on water, but remain longer when deposited on materials such as gold (as an example). This highlights the critical role of different source-target materials — a factor overlooked in more traditional Langmuir, BET, and sticking coefficient models — on desorption and activation energy. None of these models incorporate the effects of different source-target material pairings or dynamic re-structuring of various contaminants during desorption which we observed in these simulations, and future studies will continue to investigate these effects so that an improved adsorption model can be incorporated into spacecraft contamination codes and analyses.

5.1 Acknowledgments

JM acknowledges many helpful discussions with Dr. Daniel Depew, and Professors Aiichiro Nakano and Ken-ichi Nomura of the University of Southern California for their invaluable feedback to initially bring him up to speed and become familiarized with molecular dynamics. Axel Kohlmeyer and Simon Gravelle also provided many useful online resources regarding molecular dynamics simulations as well. Simulations were carried out using resources made available by USC's Center for Advanced Research Computing (CARC), and CARC Research Scientist Marco Olugin in particular was very helpful in creating custom LAMMPS builds for JM's simulations.

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